Summary of the Live Professor Q&A on Supervised/Unsupervised Learning at the 2026 Astrostatistics Summer School (06/02/2026)

1 Quick recap

This meeting was an informal discussion session during a summer school on machine learning in astronomy, where participants shared thoughts about statistical modeling and data analysis techniques. Eric discussed the use of parametric models in astronomy, including examples like the Planck function for blackbody radiation and the Schechter function for galaxy luminosities, explaining how astronomers often use heuristic approaches to find fitting formulas before understanding the underlying physics. The conversation covered challenges with unsupervised clustering methods like k-means, with Eric warning that non-parametric clustering approaches can give unreliable and unstable results, while parametric methods like Gaussian mixture models can provide more reliable results. Participants discussed supervised learning techniques including support vector machines and random forests, with Eric highlighting the rapid development and adoption of machine learning methods in astronomy. The discussion also addressed concerns about methodology reliability and the importance of being cautious about different analytical approaches, with Eric emphasizing that astronomers should be aware of potential biases and limitations in their chosen methods.

2 Summary

2.1 Parametric Modeling in Astronomy

Eric and Anaya discussed the use of parametric modeling in astronomy, particularly when there is no clear physical motivation for a specific random variable distribution. Eric explained that while physically motivated models like the Planck function are preferred, heuristic models based on fitting data to distributions like the kappa distribution or gamma distribution can still be valuable for summarizing data with a few parameters. George raised a related question about distinguishing between measurement limitations and actual process distributions in data modeling.

2.2 Astronomical Modeling Techniques Discussion

Eric discussed the challenges of modeling in astronomy, particularly addressing deterministic versus stochastic components and biases in samples, including the use of survival analysis to handle upper limits in data. He provided a critical assessment of unsupervised clustering methods, warning about their instability and recommending Gaussian mixture models as a more reliable alternative. Eric also reviewed supervised learning techniques, highlighting the importance of support vector machines and random forests, and noted their increasing adoption in astronomical research.

2.3 t-SNE Dimensionality Reduction Overview

Eric explained that t-SNE uses a Student's t-distribution with heavier tails compared to Gaussian distributions, and attributed its development to Geoffrey Hinton. He clarified that t-SNE is a dimensionality reduction tool for visualization rather than a classifier, and noted its popularity in astronomy and other fields. Eric recommended Wikipedia as a starting resource for learning about statistical methods, along with specialized textbooks and R implementations for practical application.

2.4 K-means Clustering Limitations in Astronomy

Pawan discussed the limitations of K-means clustering, particularly its struggles with centroid initialization and sensitivity to outliers, recommending it not be used by astronomers. Eric agreed with Pawan's concerns about K-means and emphasized the importance of being cautious about methodology in astronomy, suggesting that researchers should try different analytical approaches to ensure science conclusions are stable. The discussion highlighted the need for astronomers to be more aware of statistical options beyond traditional methods and to be prepared for situations where data sets may not yield reliable scientific results.

2.5 Research Reproducibility Standards Discussion

The group discussed the importance of collecting chat discussions for generating accurate summaries, with Andrew confirming he has transcripts of past meetings. Eric explained how statisticians focus on precision to ensure reliable scientific conclusions, contrasting this with scientists who may be more casual in their communication. The conversation then shifted to a discussion about reproducibility issues in research, where Eric noted that astronomical research avoided the 2017 crisis that affected biomedical and social science research by already using three-sigma standards rather than the problematic p¡0.05 threshold.

2.6 AI and Statistics in Astronomy

The discussion focused on the role of AI and statistical methodology in astronomy, with Eric emphasizing that while AI tools like Claude can be effective for writing code, astronomers must still think critically and not rely solely on automated methods. Eric and George discussed the wide range of methodological approaches used in astronomy, from outdated methods like mean and standard deviation to advanced AI techniques, highlighting the importance of statistical expertise. The conversation included guidance for young astronomers on learning statistics, with Eric encouraging them to become experts in methodology and suggesting they might outperform their professors in this area, while also recommending specific tools like R and Python for different applications.